Accounting for Long-Term Assets

Chapter 8

Wild and Shaw

Financial and Managerial Accounting

2025 Release

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Chapter 8 Learning Objectives

CONCEPTUAL

C1 Compute the cost of plant assets.
C2 Explain depreciation for partial years and changes in estimates.
C3 Distinguish between revenue and capital expenditures, and account for them.

ANALYTICAL

A1 Compute total asset turnover and apply it to analyze a company's use of assets.

PROCEDURAL

P1 Compute and record depreciation using the straight-line, units-of-production, and declining-balance methods.
P2 Account for asset disposal through discarding or selling an asset.
P3 Account for natural resource assets and their depletion.
P4 Account for intangible assets.
P5 *Appendix 8A*—Account for asset exchanges.

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Plant Assets: Definition

Tangible in Nature

↓

Actively Used in Operations

↓

Expected to Benefit Future Periods

↓

Called Property, Plant, & Equipment

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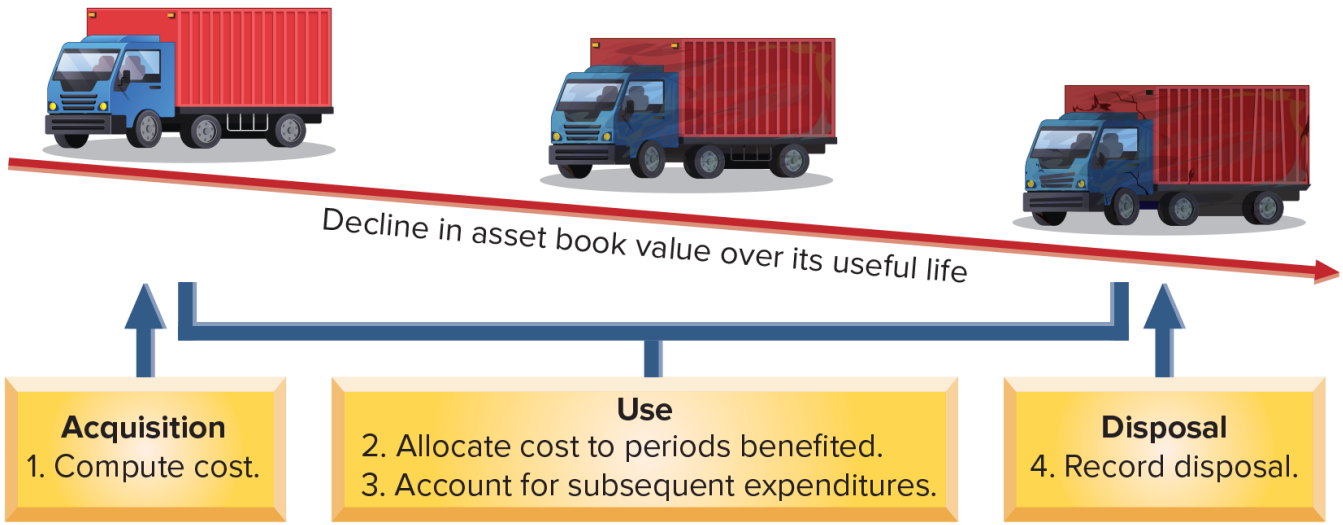
Plant Assets: Four Issues

(1) Compute the costs of plant assets
(2) Allocate the costs of plant assets
(3) Account for expenditures to plant assets
(4) Record the disposal of plant assets

Exhibit 8.2



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Learning Objective C1

Compute the cost of plant assets.

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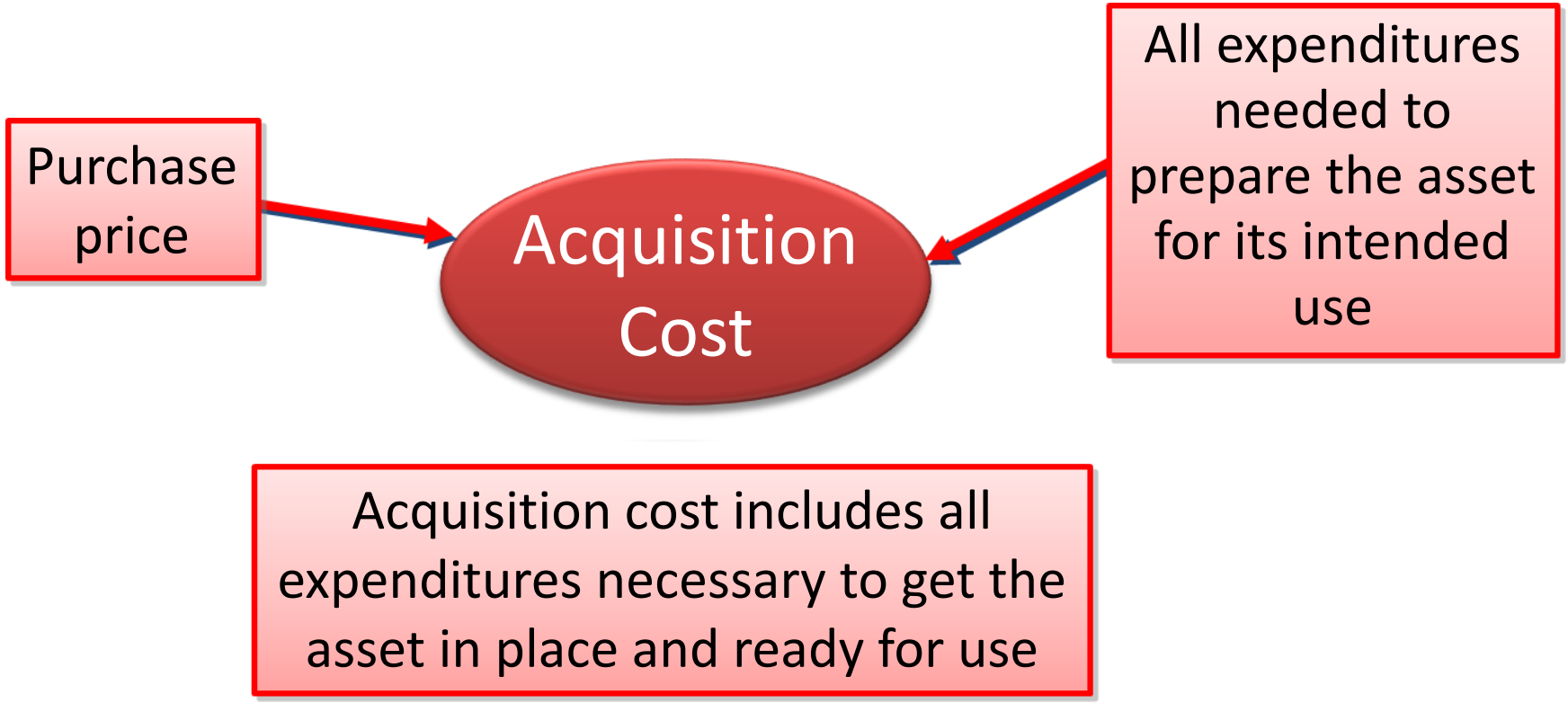
Cost Determination



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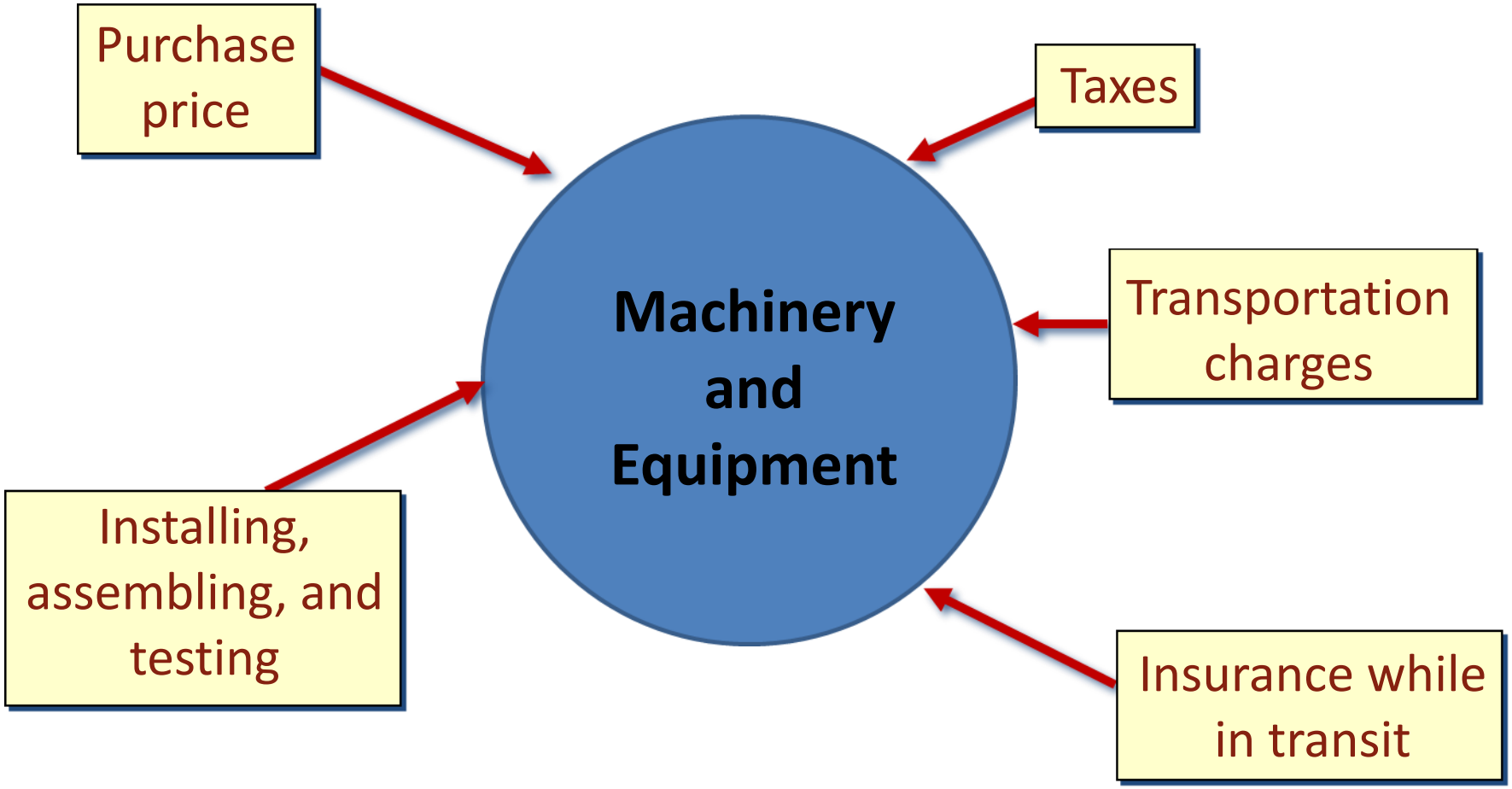
Machinery and Equipment



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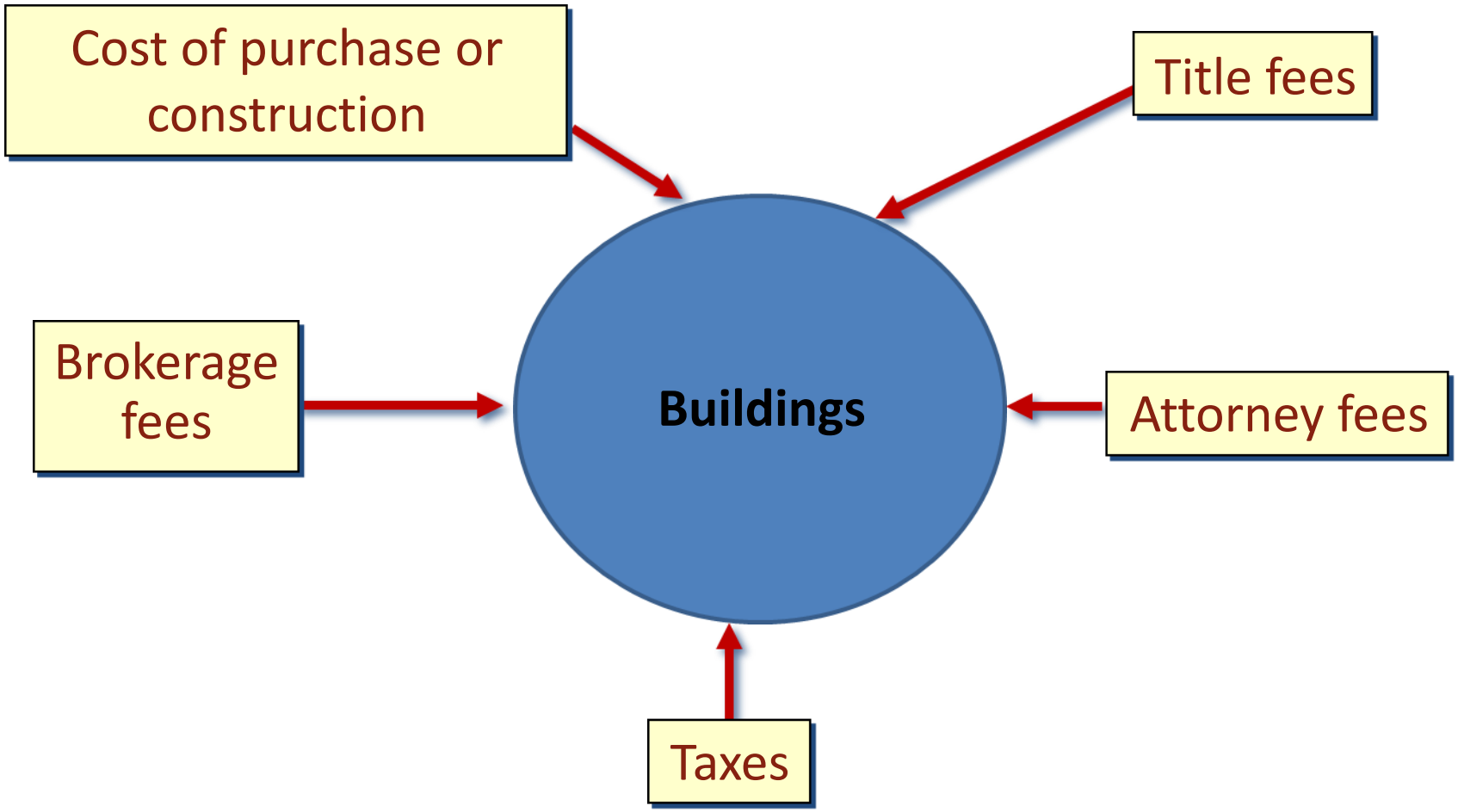
Buildings



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Land Improvements

Parking lots, driveways, fences, walkways, and lighting systems.

Depreciate over useful life of improvements.

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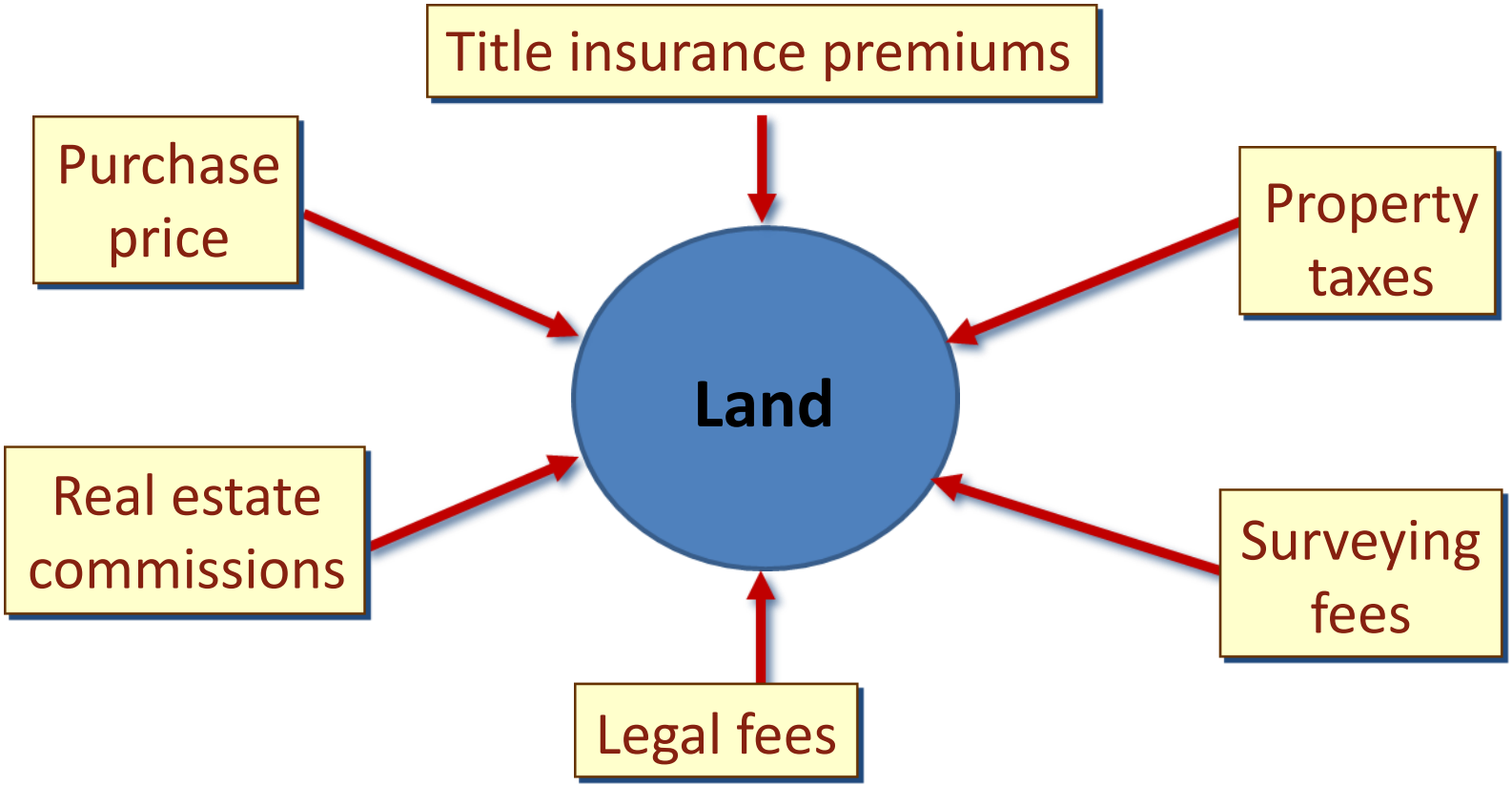
Land



Land is **not** depreciable.

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Lump-Sum Purchase

The total cost of a combined purchase of land and building is allocated based of their relative fair market values.

CarMax paid $90,000 cash to acquire a group of items consisting of land appraised at $40,000 and a building appraised at $60,000. The $90,000 cost will be allocated on the basis of appraised values as shown:

Exhibit 8.4

	Appraised Value	Percent of Total	Apportioned Cost
Building	$ 60,000	60% ($60,000/$100,000)	**$54,000** ($90,000 × 60%)
Land	40,000	40 ($40,000/$100,000)	**36,000** ($90,000 × 40%)
Totals	$100,000	100%	$90,000

Entry for lump-sum cash purchase:

Building	54,000	
Land	36,000	
Cash		90,000
Record costs of assets.		

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Learning Objective P1

Compute and record depreciation using the straight-line, units-of-production, and declining-balance methods.

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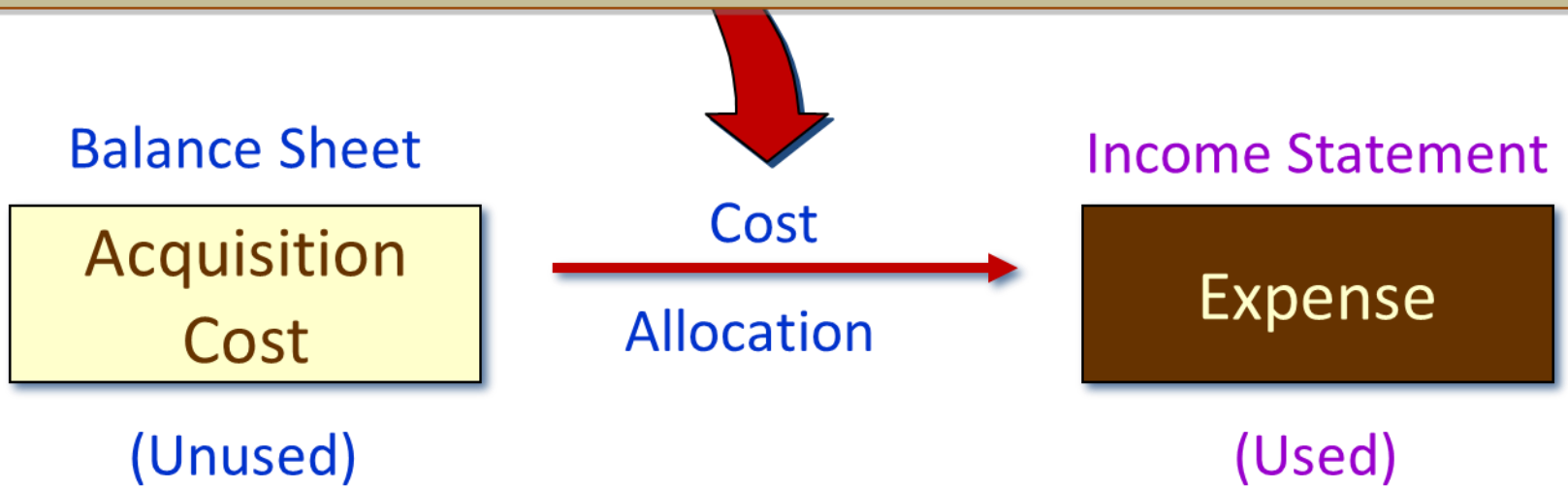
Depreciation

Depreciation is the process of allocating the cost of a plant asset to expense in the accounting periods benefiting from its use.



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Factors in Computing Depreciation

The calculation of depreciation requires three amounts for each asset:

1. **Cost**
2. **Salvage Value**
3. **Useful Life**

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Depreciation Methods

1. **Straight-line**
2. **Units-of-production**
3. **Declining-balance**

Cost	$10,000
Salvage value	1,000
Depreciable cost	$ 9,000

Useful life:	
Accounting periods	5 years
Units inspected	36,000 shoes

Exhibit 8.5

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Straight-Line Method: Example

Exhibit 8.6

$$\frac{\textbf{Cost} - \textbf{Salvage value}}{\textbf{Useful life in periods}} = \frac{\$10,000 - \$1,000}{5 \text{ years}} = \$1,800 \text{ per year}$$

Dec. 31	Depreciation Expense	1,800	
	Accumulated Depreciation—Machinery		1,800
	Record annual depreciation.		

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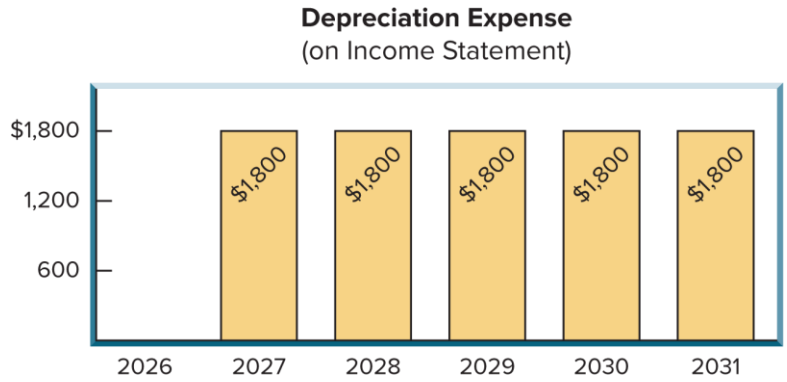
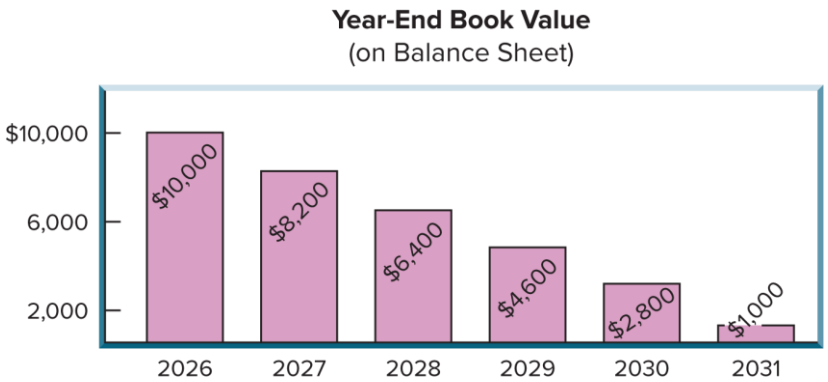
Straight-Line Method: Balance Sheet

Exhibit 8.7





Balance Sheet Presentation		
Machinery	$ 10,000	
Less: accumulated depreciation	3,600	$ 6,400

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Straight-Line Depreciation Schedule

Exhibit 8.8

	Depreciation for the Period			End of Period	
Annual Period	Depreciable Cost*	Depreciation Rate	Depreciation Expense	Accumulated Depreciation	Book Value†
2026	—	—	—	—	$10,000
2027	$9,000	20%	$1,800	$1,800	8,200
2028	9,000	20	1,800	3,600	6,400
2029	9,000	20	1,800	5,400	4,600
2030	9,000	20	1,800	7,200	2,800
2031	9,000	20	1,800	9,000	**1,000**
			$9,000	$10,000 cost – $1,000 salvage	

*$10,000 – $1,000. †Book value is total cost minus accumulated depreciation.

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Units-of-Production Method: Two-Step Process

Exhibit 8.9

Step 1:

$$\text{Depreciation Per Unit} = \frac{\text{Cost} - \text{Salvage Value}}{\text{Total Units of Production}}$$

Step 2:

$$\text{Depreciation Expense} = \text{Depreciation Per Unit} \times \text{Number of Units Produced in the Period}$$

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Units-of-Production Method: Example

Exhibit 8.9

Cost	$10,000
Salvage value	1,000
Depreciable cost	$ 9,000
Useful life	
Accounting periods	5 years
Units inspected	36,000 shoes

Assume that 7,000 units were inspected during the first year. Depreciation would be calculated as follows:

1:

$$\text{Depreciation Per Unit} = \frac{\text{Cost} - \text{Salvage Value}}{\text{Total Units of Production}} = \frac{\$9{,}000}{36{,}000} = \$0.25/\text{unit}$$

2:

$$\text{Depreciation Expense} = \text{Depreciation Per Unit} \times \text{Number of Units Produced in the Period} = \$0.25 \times 7{,}000 = \$1{,}750$$

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Units-of-Production Depreciation Schedule

Exhibit 8.10

Annual Period	Depreciation for the Period: Number of Units	Depreciation for the Period: Depreciation per Unit	Depreciation for the Period: Depreciation Expense	End of Period: Accumulated Depreciation	End of Period: Book Value
2026	—	—	—	—	$10,000
2027	7,000	$0.25	$1,750	$1,750	8,250
2028	8,000	0.25	2,000	3,750	6,250
2029	9,000	0.25	2,250	6,000	4,000
2030	7,000	0.25	1,750	7,750	2,250
2031	5,000	0.25	1,250	9,000	1,000
	36,000 units	$10,000 cost – $1,000 salvage	$9,000	Salvage value is not depreciated.	

Units produced and sold during the period.

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Declining-Balance Method: Three Steps

Cost	$10,000
Salvage value	1,000
Depreciable cost	$ 9,000
Useful life	
Accounting periods	5 years
Units inspected	36,000 shoes

Exhibit 8.11

Step 1 **Straight-line rate = 100% ÷ Useful life** = 100% ÷ 5 years = 20%

Step 2 **Double-declining-balance rate = 2 × Straight-line rate** = 2 × 20% = 40%

Step 3 **Depreciation expense = Double-declining-balance rate × Beginning-period book value**

40% × $10,000 = $4,000 (for 2027)

*In simple form: DDB depreciation = (2 × Beginning-period book value)/Useful life.

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Double-Declining-Balance Method: Schedule

Annual Period	Depreciation for the Period: Beginning-of-Period Book Value	Depreciation for the Period: Depreciation Rate	Depreciation for the Period: Depreciation Expense	End of Period: Accumulated Depreciation	End of Period: Book Value
2026	—	—	—	—	$10,000
2027	$10,000	40%	$4,000	$4,000	6,000
2028	6,000	40	2,400	6,400	3,600
2029	3,600	40	1,440	7,840	2,160
2030	2,160	40	864	8,704	1,296
2031	1,296	40	296	9,000	1,000
		$10,000 cost – $1,000 salvage	$9,000	Salvage value is not depreciated.	

Exhibit 8.12

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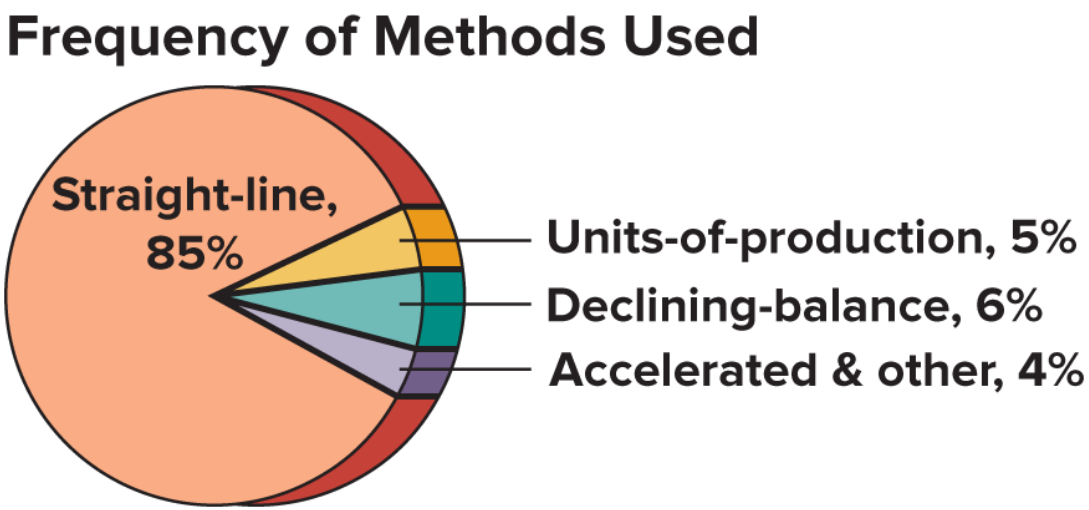
Comparing Depreciation Methods

Period	Straight-Line	Units-of-Production	Double-Declining-Balance
2027	$1,800	$1,750	$4,000
2028	1,800	2,000	2,400
2029	1,800	2,250	1,440
2030	1,800	1,750	864
2031	1,800	1,250	296
Totals	**$9,000**	**$9,000**	**$9,000**

Exhibit 8.13

Methods Used by Companies



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Depreciation for Tax Reporting

Most corporations use the Modified Accelerated Cost Recovery System (MACRS) for tax purposes.

MACRS depreciation provides for rapid write-off of an asset's cost in order to stimulate new investment.

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Learning Objective C2

Explain depreciation for partial years and changes in estimates.

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Partial-Year Depreciation

When a plant asset is purchased (or sold) during the year, depreciation is calculated for the fraction of the year the asset is owned.

Cost	$	10,000
Salvage value		1,000
Depreciable cost	$	9,000
Useful life		
Accounting periods		5 years
Units inspected		36,000 units

Assume our machinery was purchased on October 1, 2026. Let's calculate depreciation expense for 2026 assuming we use straight-line depreciation.

$$\frac{\$10,000 - \$1,000}{5 \text{ years}} \times \frac{3}{12} = \$450$$

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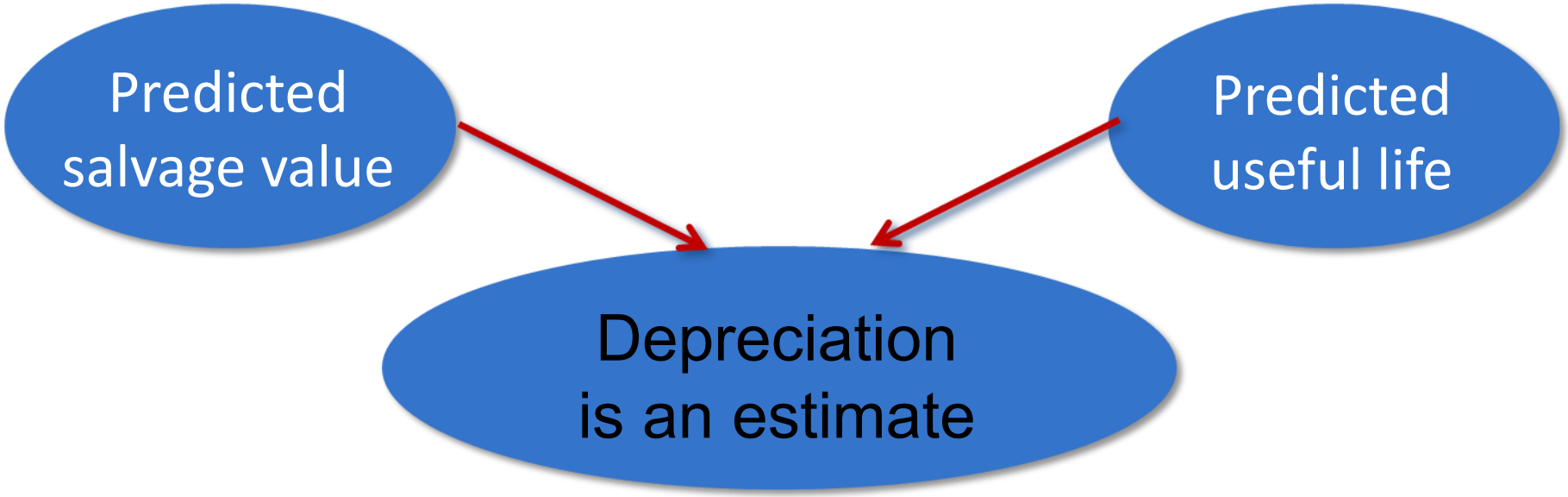
Changes in Estimates



Over the life of an asset, new information may come to light that indicates the original estimates were inaccurate.

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Changes in Estimates for Depreciation

Let's look at our machinery from the previous examples and assume that at the beginning of the asset's third year, its book value is $6,400 ($10,000 cost less $3,600 accumulated depreciation using straight-line depreciation). At that time, it is determined that the machinery will have a remaining useful life of 4 years, and the estimated salvage value will be revised downward from $1,000 to $400.

$$\frac{\textbf{Book value} - \textbf{Revised salvage value}}{\textbf{Revised remaining useful life}} = \frac{\$6,400 - \$400}{4\text{ years}} = \$1,500 \text{ per year}$$

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Learning Objective C3

Distinguish between revenue and capital expenditures, and account for them.

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Additional Expenditures

Revenue expenditures

- Do not materially increase the plant asset's life or capabilities.
- Recorded as an expense in the current period.
- Reported on the income statement.

Capital expenditures

- Provide benefits for longer than the current period.
- Recorded as an addition to the asset account.
- Reported on the balance sheet.

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Revenue and Capital Expenditures

Type of Expenditure	Capital or Revenue	Identifying Characteristics
Ordinary Repairs	Revenue	1. Maintains normal operating condition. 2. Does not increase productivity. 3. Does not extend life beyond original estimate.
Betterments and Extraordinary Repairs	Capital	1. Major overhauls or partial replacements. 2. Extends life beyond original estimate.

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Revenue and Capital Expenditures: Journal Entries

Ordinary Repairs

Dec. 31	Repairs Expense	9,500	
	Cash		9,500
	Record ordinary repairs of equipment		

Betterments (Improvements)

Jan. 2	Machinery	1,800	
	Cash		1,800
	Record installation of automated system.		

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Learning Objective P2

Account for asset disposal through discarding or selling an asset.

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Disposals of Plant Assets

Update depreciation to the date of disposal.

Journalize disposal by:

- Recording cash received (debit) or paid (credit).
- Removing accumulated depreciation (debit).
- Removing the asset cost (credit).
- Recording a gain (credit) or loss (debit).

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Discarding Plant Assets

If Cash > BV, record a gain (credit).

If Cash < BV, record a loss (debit).

If Cash = BV, no gain or loss.

Recording cash received (debit) or paid (credit).

Recording a gain (credit) or loss (debit).

Removing accumulated depreciation (debit).

Removing the asset cost (credit).

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Discarding Plant Assets: Asset Fully Depreciated

- A machine costing $9,000, with accumulated depreciation of $9,000 on December 31 of the previous year, was discarded on June 5th of the current year.
- The company is depreciating the equipment using the straight-line method over eight years with zero salvage value.

Steps 1–4:

June 5	Accumulated Depreciation—Machinery	9,000	
	Machinery		9,000
	Discarding of fully depreciated machinery.		

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Discarding Plant Assets: Asset Not Fully Depreciated

- Equipment costing $8,000, with accumulated depreciation of $6,000 on 12/31 of previous year, was discarded on 7/1.
- The company is using straight-line depreciation over eight years with zero salvage value.

Step 1: Record depreciation expense to July 1.

July 1	Depreciation Expense	500	
	Accumulated Depreciation—Equipment		500
	Record 6 months' depreciation ($1,000 × 6/12).		

Steps 2–4:

July 1	Accumulated Depreciation—Equipment	6,500	
	Loss on Disposal of Equipment	1,500	
	Equipment		8,000
	Discard equipment with a $1,500 book value.		

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Selling Plant Assets – At Book Value

- 3/31, BTO sells equipment that originally cost $16,000 and has accumulated depreciation of $12,000 at 12/31 of the prior year.
- BTO uses straight-line depreciation at $4,000 per year.
- The equipment is sold for $3,000 cash.

Step 1: Record depreciation expense to March 31.

Mar. 31	Depreciation Expense	1,000	
	Accumulated Depreciation—Equipment		1,000
	Record 3 months' depreciation ($4,000 × 3/12).		

Step 2: Record sale of asset at book value ($16,000 − $13,000 = $3,000).

Mar. 31	Cash	3,000	
	Accumulated Depreciation—Equipment	13,000	
	Equipment		16,000
	Record sale of equipment for no gain or loss.		

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Selling Plant Assets – Above Book Value

- 3/31, BTO sells equipment that originally cost $16,000 and has accumulated depreciation of $12,000 at 12/31 of the prior year.
- BTO uses straight-line depreciation at $4,000 per year.
- The equipment is sold for $7,000 cash.

Step 1: Update depreciation to March 31st.

Mar. 31	Depreciation Expense	1,000	
	Accumulated Depreciation—Equipment		1,000
	Record 3 months' depreciation ($4,000 × 3/12).		

Step 2: Record sale of asset at a gain. ($7,000 − $3,000 = $4,000 gain).

Mar. 31	Cash	7,000	
	Accumulated Depreciation—Equipment	13,000	
	Gain on Disposal of Equipment		4,000
	Equipment		16,000
	Record sale of equipment for a $4,000 gain.		

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Selling Plant Assets – Below Book Value

- 3/31, BTO sells equipment that originally cost $16,000 and has accumulated depreciation of $12,000 at 12/31 of the prior year.
- BTO uses straight-line depreciation at $4,000 per year.
- The equipment is sold for $2,500 cash.

Step 1: Update depreciation to March 31st.

Mar. 31	Depreciation Expense	1,000	
	Accumulated Depreciation—Equipment		1,000
	Record 3 months' depreciation ($4,000 × 3/12).		

Step 2: Record sale of asset at a loss. ($3,000 − $2,500 = $500 loss).

Mar. 31	Cash	2,500	
	Loss on Disposal of Equipment	500	
	Accumulated Depreciation—Equipment	13,000	
	Equipment		16,000
	Record sale of equipment for a $500 loss.		

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Learning Objective P3

Account for natural resource assets and their depletion.

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Natural Resources

Total cost is charged to depletion expense over periods benefited.

Extracted from the natural environment and reported at cost less accumulated depletion.

Examples: oil, coal, gold

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Cost Determination and Depletion

A mineral deposit with an estimated 250,000 tons of available ore is purchased for $500,000, and we expect zero salvage value. The depletion charge is $2 per ton. If 85,000 tons are mind and sold, the depletion charge is $170,000.

Exhibit 8.16

Step 1 $$\text{Depletion per unit} = \frac{\text{Cost} - \text{Salvage value}}{\text{Total units of capacity}} = \frac{\$500{,}000 - \$0}{250{,}000 \text{ tons}} = \$2 \text{ per ton}$$

Step 2 $$\text{Depletion expense} = \text{Depletion per unit} \times \text{Units extracted and sold in period}$$
$$= \$2 \times 85{,}000 \text{ tons} = \$170{,}000$$

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Depletion of Natural Resources

Depletion expense in the first year would be:

Dec. 31	Depletion Expense—Mineral Deposit	170,000	
	Accumulated Depletion—Mineral Deposit		170,000
	Record depletion of the mineral deposit.		

Balance Sheet presentation of natural resources:

Exhibit 8.17

Mineral deposit	$500,000	
Less accumulated depletion	170,000	$330,000

OR

Mineral deposit (net of $170,000 accumulated depletion)	$330,000

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Depletion of Natural Resources: Journal Entry

Depletion expense when some ore remains unsold at year-end:

Dec. 31	Depletion Expense—Mineral Deposit	140,000	
	Ore Inventory	30,000	
	Accumulated Depletion—Mineral Deposit		170,000
	Record depletion and inventory of mineral deposit.		

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Plant Assets Tied into Extracting

- Specialized plant assets may be required to extract the natural resource.
- These assets are recorded in a separate account and depreciated.

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Learning Objective P4

Account for intangible assets.

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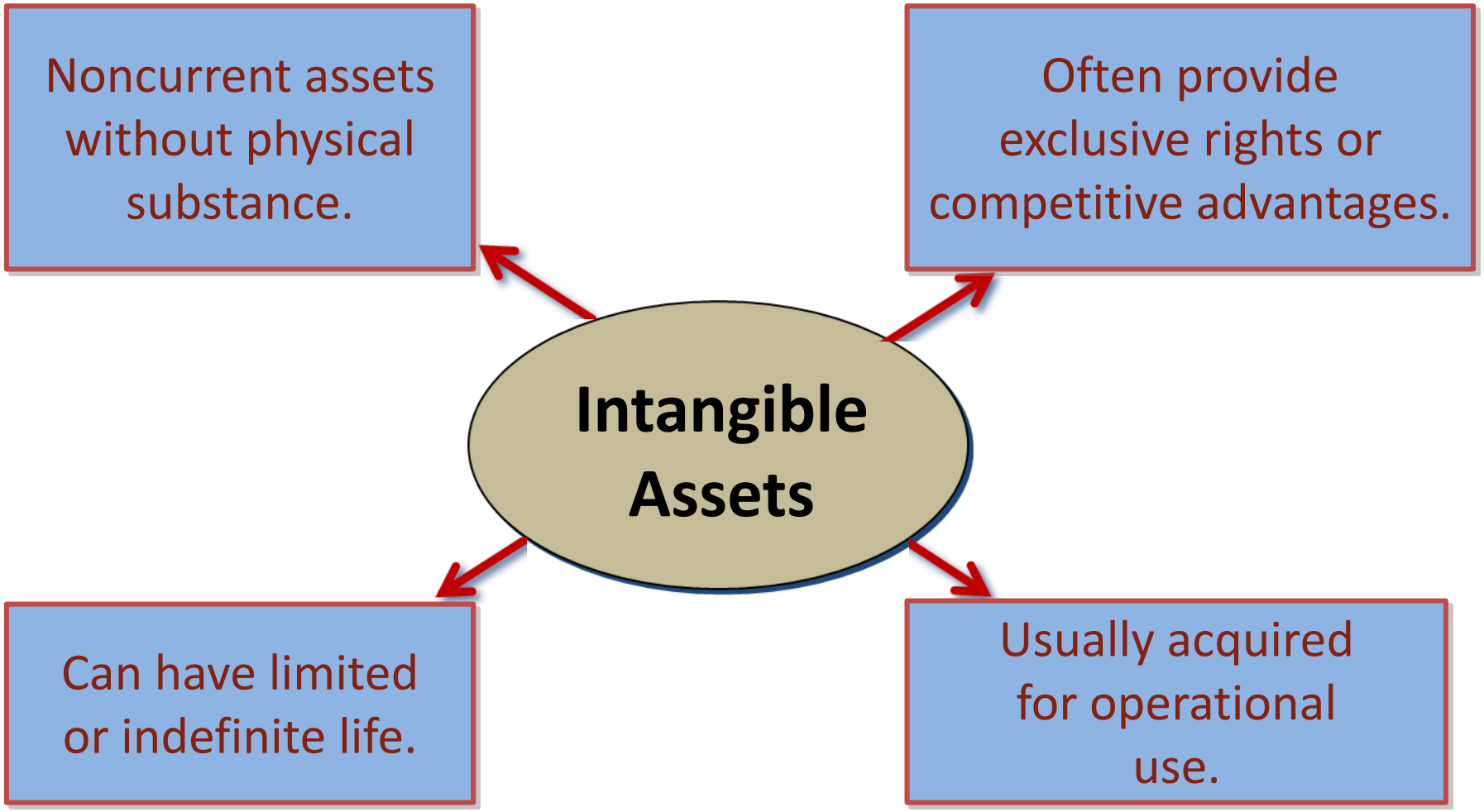
Intangible Assets



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Cost Determination and Amortization

Record at cost, including purchase price, legal fees, and filing fees.

- Patents
- Copyrights
- Franchises and Licenses
- Trademarks and Trade Names
- Goodwill
- Right-of-Use Asset (Lease)
- Leasehold Improvements
- Other Intangibles
- Research and Development

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Patents, Copyrights, Franchises, and Licenses

- Patent – owner has exclusive right to manufacture and sell a patented item or use a process for 20 years.
- Copyright – gives owner exclusive right to publish and sell a musical, literary, or artistic work during the life of the creator plus 70 years.
- Franchises and licenses – rights to sell a product or service under specified conditions.

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Goodwill and Leases

- Goodwill – amount by which company's value exceeds value of its individual assets and liabilities. Only recorded when entire company or business segment is purchased.
- Right-of-Use Asset (Lease) – property rented under a contract.
- Leasehold Improvements – improvements the lessee pays for improvements to the leased property.
- Other Intangibles – incudes software, noncompete covenants, and customer lists.

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Research and Development

Research and Development – expenditures to discover new products, processes, or knowledge.

- Costs are expensed when incurred.
- GAAP does not include them as intangible assets.

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Learning Objective A1

Compute total asset turnover and apply it to analyze a company's use of assets.

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Total Asset Turnover

$$\text{Total asset turnover} = \frac{\text{Net sales}}{\text{Average total assets}}$$

Exhibit 8.18

Provides information about a company's efficiency in using its assets.

Company	Figure ($ millions)	Current Year	1 Year Ago	2 Years Ago
Starbucks	Net sales	$35,976	$32,250	$29,061
	Average total assets	$28,712	$29,686	$30,384
	Total asset turnover	**1.25**	**1.09**	**0.96**
Jack in the Box	Net sales	$ 1,692	$ 1,468	$ 1,144
	Average total assets	$ 2,962	$ 2,336	$ 1,828
	Total asset turnover	**0.57**	**0.63**	**0.63**

Exhibit 8.19

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Learning Objective P5

Appendix 8A

Account for asset exchanges.

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Exchanging Plant Assets

- Many plant assets such as machinery, automobiles, and office equipment are exchanged for newer assets.
- In an exchange, a trade-in allowance is received on the old asset and the balance is paid in cash.
- Accounting for the exchange of assets depends on whether the transaction has commercial substance.

Commercial substance implies the company's future cash flows will be altered.

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Exchange with Commercial Substance

Exhibit 8A.1

- A company acquires $42,000 in new equipment.
- In exchange, the company pays $33,000 cash and trades in old equipment.
- The old equipment originally cost $36,000 and has accumulated depreciation of $20,000 (book value is $16,000).
- This exchange has commercial substance. The old equipment has a trade-in allowance of $9,000.

Asset Exchange Has Commercial Substance	Loss	Loss	Gain	Gain
Market value of asset received		$42,000		$42,000
Book value of assets given:				
Equipment ($36,000 – $20,000)	$16,000		$16,000	
Cash	33,000	49,000	23,000	39,000
Gain (loss) on exchange		**$ (7,000)**		**$ 3,000**

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Exchange with Commercial Substance: Journal Entry for a Loss

- A company acquires $42,000 in new equipment.
- In exchange, the company pays $33,000 cash and trades in old equipment.
- The old equipment originally cost $36,000 and has accumulated depreciation of $20,000 (book value is $16,000).
- This exchange has commercial substance. The old equipment has a trade-in allowance of $9,000.

Jan. 3	Equipment (**new**)	42,000	
	Loss on Exchange of Assets	7,000	
	Accumulated Depreciation—Equipment (**old**)	20,000	
	Equipment (**old**)		36,000
	Cash		33,000
	Record exchange (with commercial substance) of old equipment and cash for new equipment.		

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Exchange with Commercial Substance: Journal Entry for a Gain

Let's assume the same facts as on the previous screen except that the company pays $23,000 cash with the trade-in.

Jan. 3	Equipment (**new**)	42,000	
	Accumulated Depreciation—Equipment (**old**)	20,000	
	Equipment (**old**)		36,000
	Cash		23,000
	Gain on Exchange of Assets		3,000
	Record exchange (with commercial substance) of old equipment and cash for new equipment.		

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